
Migrating from ST25R95 to ST25R200 multipurpose NFC transceiver

Introduction

The ST25R200 transceiver delivers high-end performance in a small 4 × 4 mm package. It brings the convenience of contactless interaction and features to a variety of end applications. Specifically designed for consumer and industrial end products, this device is a versatile solution for enabling NFC use cases. With its compact size and advanced features, it is an ideal choice for manufacturers looking to incorporate NFC technology in their products.

The purpose of this document is to provide a list of considerations and requirements to ensure a successful migration from ST25R95 to ST25R200. The document is based on the ST25R200 device, but its content applies to ST25R100 as well.

For further information, refer to the documents listed below:

- DS12807: ST25R95 datasheet
- DS13658: ST25R200 datasheet
- DS14139: ST25R100 datasheet
- AN5984: *Layout recommendations for the design of boards with ST25R200*
- UM2890: *RF/NFC abstraction layer (RFAL)*
- AN6065: *Antenna design for ST25R200 device*
- AN5993: *Wake-up mode for the ST25R200*

1 Terminology

The table below defines acronyms and other terms used in this document.

Table 1. Acronyms

Acronym	Description
ACM	Active communication mode
ADC	Analog-to-digital converter
AFE	Analog front end
GPT	General purpose timer
HAL	Hardware abstraction layer
IRQ	Interrupt request
MCU	Microcontroller unit
MRT	Mask receive timer
NRT	No response timer
P2P	Peer to peer
PCB	Printed circuit board
PCM	Passive communication mode
PSSR	Power supply rejection ratio
RF	Radio frequency
RFAL	RF abstraction layer
RW	Read/Write
WU	Wake-up
WUT	Wake-up timer

2 Signal and pinout comparison

ST25R95 and ST25R200 devices are available in QFN packages. ST25R95 devices use a 5 x 5 mm 32-pin package, ST25R200 devices feature a smaller (4 x 4 mm) 24-pin package.

The pinouts are not pin-to-pin compatible.

2.1 Signal layout comparison

The figures below show the pinouts of the two devices, for signal layout comparison.

Table 2. Comparison between ST25R95 and ST25R200

ST25R95	ST25R200
Note: Shaded area represents the dissipation pad (it must be connected to ground).	

2.2 Pinout differences

The table below describes the main pinout differences between the ST25R95 and the ST25R200 devices.

Table 3. Pinout comparison

ST25R95			ST25R200	
Pin	Name	Description	Name	Description
1	TX1	Driver output 1	IRQ	Interrupt request output
2	TX2	Driver output 2	MISO	SPI slave output
3	NC	Not connected	VDD_IO	External peripheral communication supply
4	NC	Not connected	GND_D	Digital ground
5	RX1	Receiver input 1	VDD_D	Regulated supply for digital
6	RX2	Receiver input 2	VDD_AM	Regulated driver supply for AM modulation
7	NC	Not connected	VDD	External main supply
8	GND_RX	Analog ground	VDD_DR	Regulated driver supply for antenna driver
9	ST_R0	Reserved	RFO1	Driver output 1
10	NC	Not connected	RFO2	Driver output 2
11	NC	Not connected	GND_DR	Antenna driver ground
12	IRQ_IN	Interrupt input	TAD2 ⁽¹⁾	Analog/digital test pin

ST25R95			ST25R200	
Pin	Name	Description	Name	Description
13	VPS	Main power supply	TAD1 ⁽¹⁾	Analog/digital test pin
14	IRQ_OUT	Interrupt output	V _{DD_A}	Regulated supply for analog
15	SPI_SS	SPI slave select (active low)	AGD	Analog reference voltage
16	SPI_MISO	SPI slave output	RFI1	Receiver input 1
17	SPI_MOSI	SPI slave input	RFI2	Receiver input 2
18	SPI_SCK	SPI serial clock	GND_A	Analog ground
19	SSI_0	Select serial communication interface	XTI	Crystal oscillator input
20	SSI_1	Select serial communication interface	XTO	Crystal oscillator output
21	ST_R1	Reserved	RESET ⁽²⁾	Reset input
22	GND	Digital ground	BSS	SPI slave select (active low)
23	NC	Not connected	MOSI	SPI slave input
24	NC	Not connected	SCLK	SPI serial clock
25	NC	Not connected	V _{SS} ⁽³⁾	Ground via exposed die pad
26	NC	Not connected	-	-
27	NC	Not connected	-	-
28	NC	Not connected	-	-
29	XIN	Crystal oscillator input	-	-
30	XOUT	Crystal oscillator output	-	-
31	GND_TX	Ground (RF drivers)	-	-
32	VPS_TX	Power supply (RF drivers)	-	-

1. The ST25R200 features a pair of dedicated pins for test/debug, which can be connected to test points
2. The ST25R200 features a reset pin (active high) able to perform a reset via an external pin. This pin can be used to put the device in the lowest power consumption mode when the NFC functionality is not needed. This pin is optional and must be tied to GND if not used. The device can also be reset via direct command, as the previous ST25R39xx devices.
3. On ST25R200, pin 25 (exposed pad) is not only used for thermal release, but also to connect to GND. A good connection to the PCB ground plane via multiple vias is extremely important to ensure proper operation.

3 Features description and comparison

3.1 Power management

The ST25R200 has two positive supply pins: V_{DD} and V_{DD_IO} .

V_{DD} is the main power supply pin. It supplies the analog and digital block through two regulators (V_{DD_A} , V_{DD_D}), as well as the transmitter supplies (V_{DD_DR} , V_{DD_AM}). The supported V_{DD} supply range goes from 2.7 to 5.5 V.

The regulated voltage is automatically adjusted to have the maximum possible regulated voltage while still having a good PSRR. All regulator pins (except V_{DD_AM}) also have corresponding negative supply pins, which are externally connected to the ground potential (V_{SS}). The recommended blocking capacitors for supply pins are:

- V_{DD} , V_{DD_DR} , V_{DD_D} , V_{DD_AM} (pins 5, 6, 7, 8): 2.2 μ F in parallel with 10 nF.
- AGD, V_{DD_IO} (pins 3, 15): 1 μ F in parallel with 10 nF.

The regulation of V_{DD_DR} is set by a defined voltage configuration or by a target voltage drop in reference to the voltage supplied at V_{DD} .

A power-down support block is introduced to maintain low power consumption and support V_{DD_D} in PD mode where the regulators are disabled.

For further information, refer to the datasheet DS13658.

Figure 1. System diagram (differential antenna driving)

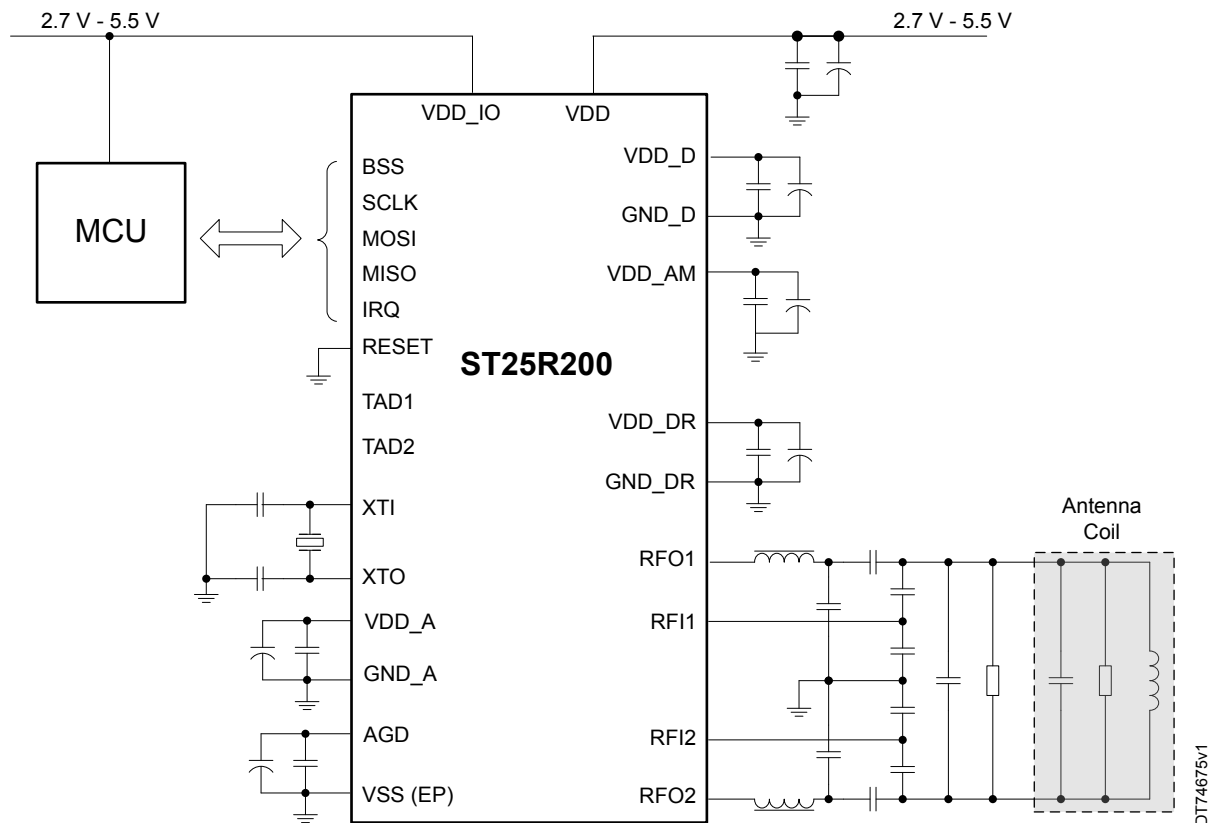
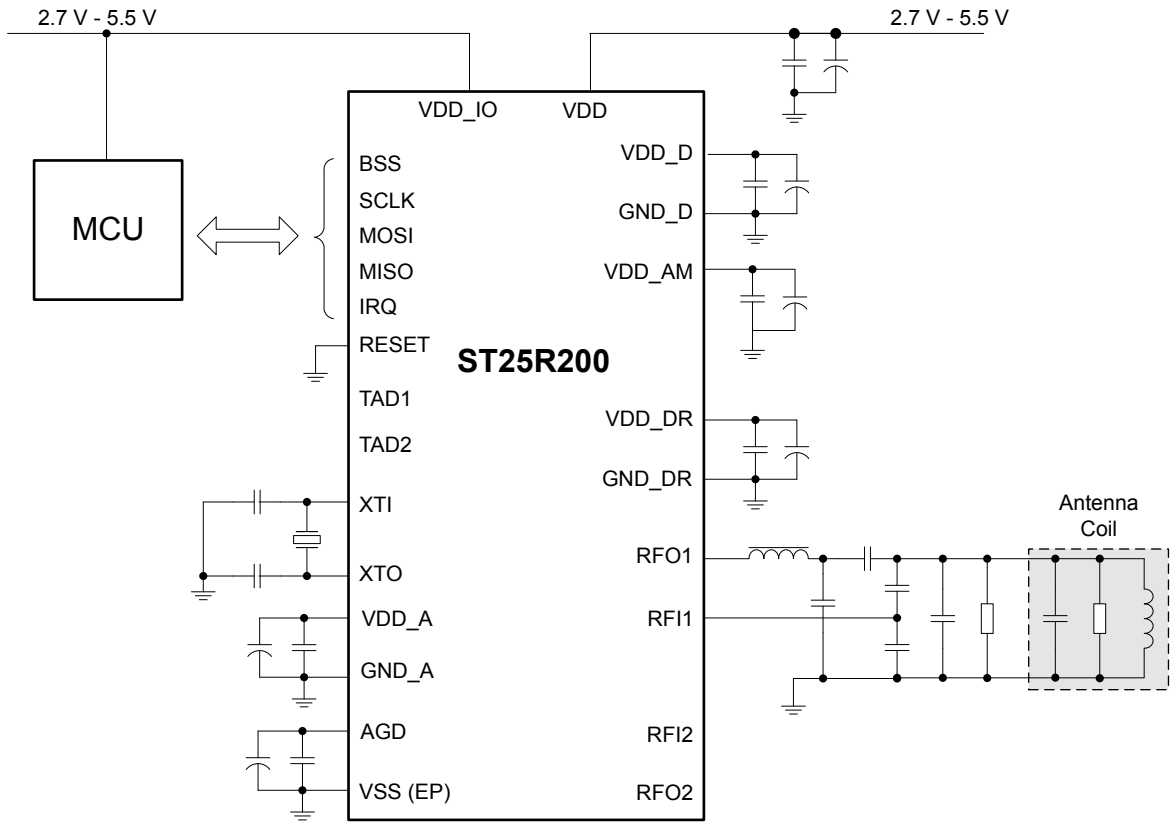


Figure 2. System diagram (single-ended driving)



DT74676v1

3.2 Serial interface

The ST25R200 uses SPI as the interface to the host, supporting speeds of up to 10 Mbit/s (6 Mbit/s on the ST25R100). Any system communicating with ST25R95 (speed up to 2 Mbit/s) via SPI is able to directly interface with ST25R200 (maintaining SPI mode and speed).

The MISO output is usually in tristate on the ST25R200, and it is driven only when output data are available. The MOSI and the MISO can be externally shorted to provide a bidirectional signal. When MISO output is in tristate, it is possible to switch on a 10 kΩ pull-down, by activating option bits `miso_pd1` and `miso_pd2`.

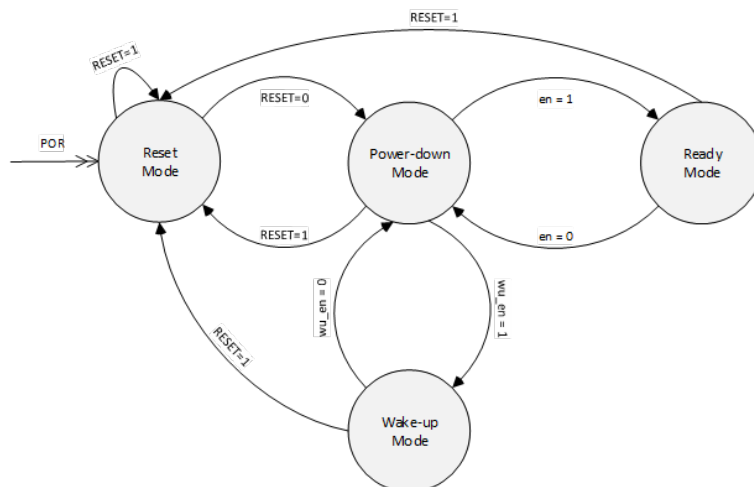
The ST25R200 does not have a UART as an interface to the host. The ST25R95 has both SPI and UART.

3.3 Operating modes

The ST25R200 has four operating modes:

- **Reset mode (RESET):** all blocks are deactivated, no clock is present, and consumption is minimized.
- **Power-down mode (PD):** The AFE static power consumption is very low, but SPI communication is possible as well as register access to those in the PD domain.
- **Wake-up mode (WU):** the device checks for a change in antenna properties in regular time intervals and triggers an IRQ in case a change was detected.
- **Ready mode (RD):** the regulators and crystal oscillator are enabled, and the device is ready to transmit/receive.

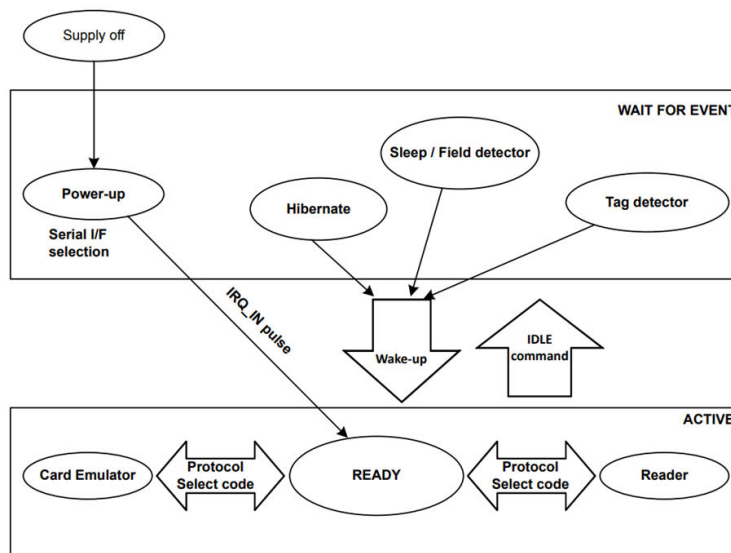
Figure 3. ST25R200 operating modes



The ST25R95 has two operating modes:

- **Wait for event (WFE)**
 - Power-up
 - Hibernate
 - Sleep/Field detector
 - Tag detector
- **Active:** after the startup sequence the ST25R95 is active and communicates with the external host and with an NFC tag.

Figure 4. ST25R95 operating modes



3.4 Consumption comparison

The following table compares the average consumption of the available modes of the ST25R95 and the ST25R200.

Table 4. Power consumption comparison

ST25R95			ST25R200	
Mode	State	Typ. ⁽¹⁾	Mode	Typ. ⁽²⁾
Wait for event	Power-up	200 μ A	Power-down	1.2 μ A
	Hibernate	1 μ A	Reset	0.12 μ A
	Sleep	20 μ A	Wake-up	1.7 μ A
	Tag detector	50 μ A	-	⁽³⁾
Active	Ready	2.5 mA	Ready	4.2 mA
	Card emulation	2.5 mA	-	-

1. ST25R95 typical values, V_{PS_Main} from 2.7 to 3.3 V.

2. ST25R200 typical values, $V_{DD} = 2.7$ V.

3. For further information, refer to AN5993.

Note: The functionality of the ST25R200 modes is similar to that of ST25R95, but the their configuration is different. For further information, refer to the DS13658.

3.5 Tag detection

ST25R200 introduces changes to the tag detection, referred as wake-up mode.

The device performs the low power card detection step where the device waits for a change in the reader environment produced by a detuning element. Once a change is detected, an interrupt is sent to the host and the device that performs the regular RF/NFC polling afterwards.

Both ST25R200 and ST25R95 devices offer a wake-up period with different interval ranges.

In addition, the ST25R200 introduces an optional autoaveraging feature that recalculates the reference value after each measurement.

3.6 Matching circuit

Different antenna matching characteristics must be considered for designs based on ST25R95 and ST25R200 devices.

The steps for designing the matching circuit for the ST25R200 are similar to those of other ST25R products.

When designing the matching circuit, be careful not to exceed the device parameters. It is recommended to use the antenna matching tool to calculate the tuning components, and to follow the procedure detailed in AN6065.

3.7 Supported NFC technologies and modes

ST25R200 does not support:

- NFC-F / FeliCa™ reader mode
- Card emulation
- P2P (PCM)

For existing projects that require the use of the above modes, migration to an ST25R39xx device is recommended.

For further details on these products, refer to the existing documentation.

4 Operation and interface

The ST25R200 has operating modes and interfaces similar to those of previous ST25R multipurpose NFC transceivers. Despite the similarities, each device has its own specificities and feature set.

The use of the STMicroelectronics NFC Library (RFAL) is recommended. RFAL supports all required technologies and protocols while encapsulating all existing ST25R devices (ST25R3911B, ST25R3916, ST25R95, ST25R200). It is not recommended to modify existing obsolete ST25R drivers.

If RFAL is already used in the project, the process of migrating from ST25R95 to ST25R200 is as simple as switching the drivers to compile.

4.1 Interface scheme

The ST25R200 scheme allows for:

- Up to 95 addressable registers
- 64 command codes (0x60 - 0x7F and 0xE0 - 0xFF)

For further information, refer to the DS13658.

4.2 Registers

There are two types of registers available in the ST25R200:

- Configuration registers
- Display registers

Configuration registers can be read and written (RW) via the SPI and contain the device configuration.

Display registers are read only (RO) and contain information about the internal state of the device.

In addition, the registers are grouped into two different power domains:

- Power down (PD)
- Ready (RD)

4.3 FIFO

On the ST25R200 reading data from the FIFO is like reading data from the addressable registers. Read and write operations are performed by accessing the address 0x5F, resulting in 0x5F for a write operation and 0xDF for a read operation.

The ST25R95 can transmit frames of up to 252 bytes, and reception of up to 528 bytes depending on the long frame support.

The ST25R200 has a FIFO of 256 bytes with the water level mechanism fixed at 128 bytes in both directions (transmission and reception). The FIFO can be reloaded during active transmission and retrieved during active reception, triggered by the water level IRQ.

No limit exists on a single frame size during reception, and transmission is restricted to 8 KB

4.4 Interrupts

The ST25R95 has a simple interrupt function with two pins, namely IRQ_IN and IRQ_OUT.

ST25R200 has an IRQ pin to notify the host, and contains three interrupt registers holding IRQ information. For further details, refer to the DS13658.

4.5 Timers

Several timers with their associated configuration registers are integrated in the ST25R200 to eliminate the need to runtime critical tasks:

- Mask receive timer (MRT)
- No-response timer (NRT)
- General purpose timer (GPT)
- Wake-up timer (WUT)

5 Migrating existing software projects

The use of the STMicroelectronics NFC library (RF abstraction Layer) on the ST25R200 is strongly recommended.

RFAL supports all required technologies and protocols while encapsulating existing ST25Rxxx devices (such as ST25R3911, ST25R3916, ST25R95, and ST25R200).

The library is regularly updated to introduce new features, wider NFC device support, and bug fixes. Compliance with the latest versions of the relevant standards is continuously ensured.

Patching and customization of existing non-RFAL drivers is not recommended. An incomplete, unsupported driver does not receive official updates.

5.1 RFAL projects

The RF Abstraction Layer (RFAL) is designed to encapsulate the ST25R device underneath.

Migration to an existing ST25R200 (or any previous ST25R) project requires only the modification of the RFAL HAL to ST25R200:

1. In the project compilation list or makefile, replace the preprocessor instruction defining the target ST25R95 device with ST25R200.
2. Replace the included folder from `rfal/source/st25r95` to `rfal/source/st25r200`.
3. Replace the relevant modules from ST25R95 with the equivalent for ST25R200.

Table 5. Modules

ST25R95	ST25R200
<code>rfal_rfst25r95</code>	<code>rfal_rfst25r200</code>
<code>st25r95</code>	<code>st25r200</code>
<code>st25r95_com</code>	<code>st25r200_com</code>
<code>st25r95_com_ce</code>	-
<code>st25r95_com_spi</code>	-
-	<code>st25r200_irq</code>

5.2 Non-RFAL projects

The standard procedure is to use the RFAL. To adapt your project accordingly, refer to the existing documentation and reference designs available on www.st.com.

6 Migrating from an existing demonstration

6.1 X-NUCLEO-NFC03A1 shield

A Nucleo shield equivalent to X-NUCLEO-NFC03A1 is available for the ST25R200. Refer to the X-NUCLEO-NFC09A1 or X-NUCLEO-NFC10A1 boards.

Table 6. Pinout comparison

Arduino pin		X-NUCLEO-NFC03A1	X-NUCLEO-NFC10A1
CN9 connector	D15	-	-
	D14	-	-
	AREF	-	-
	GND	GND	-
	D13	SPI CLK	SPI CLK
	D12	SPI MISO	SPI MISO
	D11	SPI MOSI	SPI MOSI
	D10	SPI CS	SPI CS
	D9	Interface pin (SSI_0)	-
	D8	UART_TX/IRQ_IN	RESET
	D7	MCU_LED1	LED4
	D6	MCU_LED2	LED5
	D5	MCU_LED3	LED6
	D4	MCU_LED4	-
	D3	-	-
	D2	UART_RX/IRQ_OUT	-
	D1	-	-
	D0	-	-
CN8 connector	A0	-	IRQ
	A1	-	LED1
	A2	-	LED2
	A3	-	LED3
	A4	-	-
	A5	-	-

Projects using PLUG-CR95HF-B boards must migrate to X-NUCLEO-NFC10A1.

6.2 Other demonstrations

Projects using other ST25R95 boards (EVAL-ST95HF or MB1054) must migrate to the STEVAL-25R200 kit, which allows easy prototyping with multiple antenna configurations, and also provides an interface to an easy-to-use GUI.

Integration with the ST software development kit for ST25 tags and dynamic tags (STSW-ST25SDK001) is also available.

Revision history

Table 7. Document revision history

Date	Version	Changes
29-Jul-2024	1	Initial release.

Contents

1	Terminology	2
2	Signal and pinout comparison	3
2.1	Signal layout comparison	3
2.2	Pinout differences	3
3	Features description and comparison	5
3.1	Power management	5
3.2	Serial interface	6
3.3	Operating modes	6
3.4	Consumption comparison	8
3.5	Tag detection	8
3.6	Matching circuit	8
3.7	Supported NFC technologies and modes	8
4	Operation and interface	9
4.1	Interface scheme	9
4.2	Registers	9
4.3	FIFO	9
4.4	Interrupts	9
4.5	Timers	9
5	Migrating existing software projects	10
5.1	RFAL projects	10
5.2	Non-RFAL projects	10
6	Migrating from an existing demonstration	11
6.1	X-NUCLEO-NFC03A1 shield	11
6.2	Other demonstrations	11
	Revision history	12
	List of tables	14
	List of figures	15

List of tables

Table 1.	Acronyms	2
Table 2.	Comparison between ST25R95 and ST25R200	3
Table 3.	Pinout comparison	3
Table 4.	Power consumption comparison	8
Table 5.	Modules	10
Table 6.	Pinout comparison	11
Table 7.	Document revision history	12

List of figures

Figure 1.	System diagram (differential antenna driving)	5
Figure 2.	System diagram (single-ended driving)	6
Figure 3.	ST25R200 operating modes.	7
Figure 4.	ST25R95 operating modes.	7

IMPORTANT NOTICE – READ CAREFULLY

STMicroelectronics NV and its subsidiaries ("ST") reserve the right to make changes, corrections, enhancements, modifications, and improvements to ST products and/or to this document at any time without notice. Purchasers should obtain the latest relevant information on ST products before placing orders. ST products are sold pursuant to ST's terms and conditions of sale in place at the time of order acknowledgment.

Purchasers are solely responsible for the choice, selection, and use of ST products and ST assumes no liability for application assistance or the design of purchasers' products.

No license, express or implied, to any intellectual property right is granted by ST herein.

Resale of ST products with provisions different from the information set forth herein shall void any warranty granted by ST for such product.

ST and the ST logo are trademarks of ST. For additional information about ST trademarks, refer to www.st.com/trademarks. All other product or service names are the property of their respective owners.

Information in this document supersedes and replaces information previously supplied in any prior versions of this document.

© 2024 STMicroelectronics – All rights reserved